BrineWatch

Repeatable, uncertainty-aware screening of desalination brine plumes with a low-cost ROV

Prototypes for Humanity 2026

Current stage: Initial Prototype or Model

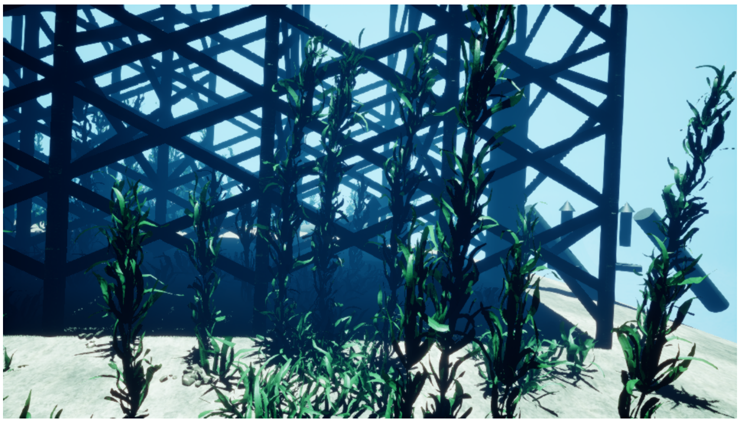
1 The problem

Desalination supplies fresh water to hundreds of millions of people and returns a dense, salty by-product to the sea. Because brine is heavier than seawater it sinks and spreads along the seabed as a thin, invisible layer — exactly where routine monitoring is weakest. Vessel casts sample a handful of points a few times per year; fixed stations rarely sit in the bottom layer. Regulators enforce mixing-zone limits that, in practice, nobody can verify continuously; operators get no feedback; communities that depend on benthic ecosystems get no evidence. Robotic plume mapping has been shown to be *possible* in research settings; BrineWatch focuses on making it **repeatable, adaptive, uncertainty-aware and affordable** — a retrofit for a BlueROV2-class mini-ROV that many institutions already own.

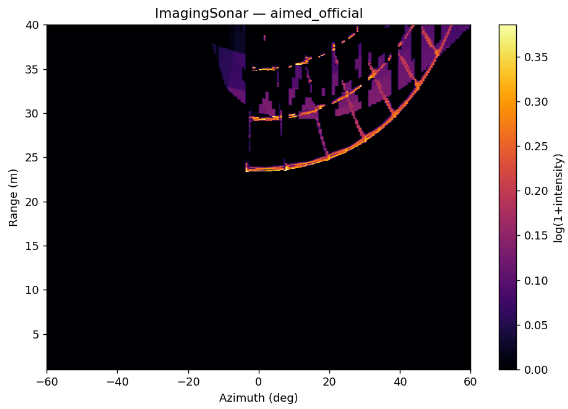
2 The prototype at a glance

Locate → Sense → Adapt → Reconstruct → Screen

sonar contacts → near-bed CT samples → boundary-aware planning → GP map + uncertainty → CLEAR / REVIEW / POSSIBLE EXCEEDANCE



[Genuine official-HoloOcean camera frame — the prototype's survey site: the outfall hardware (seabed pipe, riser with nozzle) built by the terrain-calibrated scene generator at the foot of stock harbor structures. The plume is an analytic overlay, never rendered water.]



[Genuine official ImagingSonar frame of the same structures: bright piling returns with acoustic shadows. The open-water control frame is zero everywhere.]

What exists today. A complete autonomous mission stack running end-to-end in the *official, unmodified* HoloOcean marine simulator with its stock BlueROV2: sonar-based outfall localization from an approximate chart position — in the official demonstration mission the structure was localized with a **6.4 m error from a 17 m chart prior** after ∼110 m of search, with no access to the true position — terrain-calibrated navigation over uneven seabed, near-bed conductivity–temperature sampling of a controlled analytic plume surrogate, Gaussian-process reconstruction with explicit uncertainty, three-state screening (correct POSSIBLE-EXCEEDANCE call on the demonstration scenario), and a self-contained digital mission report. The team owns the target hardware (BlueROV2 + Omniscan SS450 side-scan sonar); the physical CT retrofit is specified with a written one-day validation protocol.

3 How it works

3.1 Official simulator, honest boundaries

Everything runs on standard HoloOcean 2.3.0 + the official Ocean package — no engine modifications. Water properties do not exist in the simulator, so the CT payload samples a documented *analytic surrogate* of a negatively buoyant discharge (near-field rise and collapse, diluting bottom-hugging gravity current, tidal advection). The surrogate is not CFD; its role is to provide exact, seedable ground truth so sampling strategies and reconstructions can be scored quantitatively.

3.2 Sonar localization without privileged knowledge

A controlled experiment proved that runtime-spawned objects are invisible to the official acoustic pipeline (bit-identical frames with/without them), so the acoustic target is genuine stock harbor structure and the spawned outfall geometry is visual dressing — stated as such everywhere. The detector is classical and reproducible: per-range-row robust background subtraction, CFAR-like thresholding, component extraction, range/bearing centroids. Measured on 538 recorded noisy mission frames, single-ping intensity does *not* separate structure from clutter — so localization relies on **spatial persistence**: world-frame contact estimates accumulated across viewpoints, mode-cluster consensus, aspect-diversity and chart-plausibility gates. Ground truth is used only by the post-mission evaluator. Any fallback to the chart prior is loudly flagged in the report and disqualifies the run as sonar evidence.

3.3 Boundary-aware adaptive sampling

After identical LOCATE and BASELINE phases, the survey planner chooses the next waypoint by a *boundary-aware Gaussian-process acquisition*: a weighted combination of posterior uncertainty, proximity to the regulatory isohaline, travel and turning cost (this is a heuristic acquisition, not formal expected information gain). A fixed lawnmower is kept as the benchmark baseline and as the routine full-coverage mode.

3.4 Uncertainty-aware screening, not certification

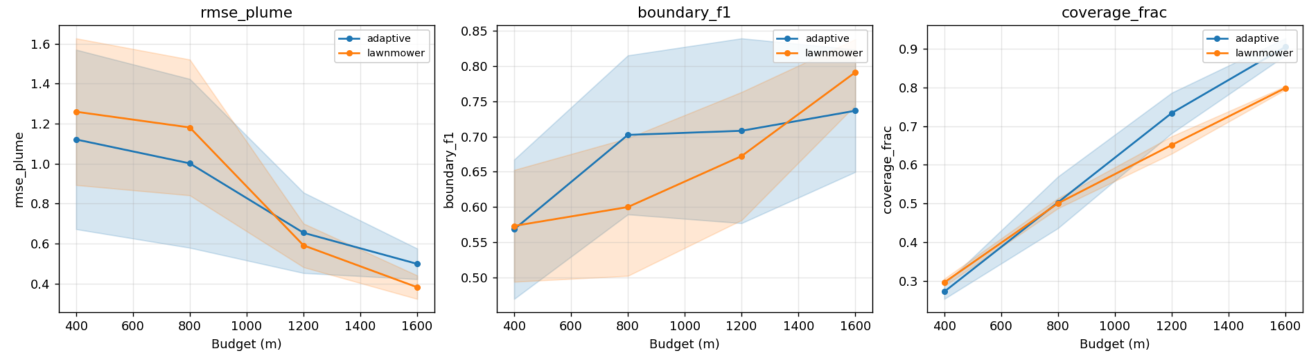
The GP posterior yields a near-bed salinity map *and* its uncertainty. The mission verdict is three-state: **CLEAR** requires both a low worst-case exceedance probability outside the mixing zone *and* low posterior uncertainty there — an unsurveyed area can never be declared compliant; **REVIEW** means the data cannot support a conclusion (recommended action: re-survey); **POSSIBLE EXCEEDANCE** escalates to accredited monitoring. Screening thresholds are explicit configuration recorded in every report.

3.5 Implemented vs planned

Implemented and exercised (simulation)	Planned (physical transfer)
mission autonomy, terrain calibration, sonar pipeline, GP mapping, screening, digital reports, benchmarks, 86 automated tests	MAVLink/ArduSub backend (backend swap by design); CT payload retrofit (BOM specified); Omniscan integration via the same record/replay interface; one-day controlled water test; harbour trial

4 Validation evidence

4.1 Equal-budget planner benchmark (20 held-out seeds per family)



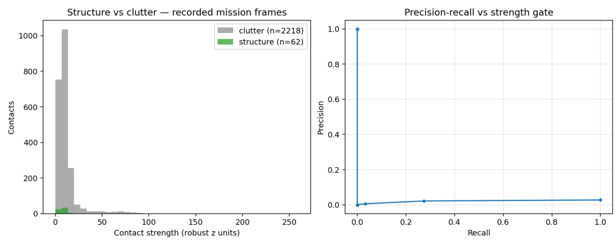
[Static primary benchmark: reconstruction error, boundary F1 and coverage vs travel budget; mean $\pm$ 1σ over 20 held-out seeds, identical LOCATE/BASELINE, equal budget.]

Budget	RMSE lawnm.	RMSE adaptive
25 %	1.26 $\pm$ 0.37	**1.12 $\pm$ 0.45**
50 %	1.18 $\pm$ 0.34	**1.00 $\pm$ 0.42**
75 %	**0.59 $\pm$ 0.11**	0.65 $\pm$ 0.20
100 %	**0.38 $\pm$ 0.06**	0.50 $\pm$ 0.08

In-plume RMSE (PSU). Boundary F1 at half budget: 0.70 (adaptive) vs 0.60 (lawnmower).

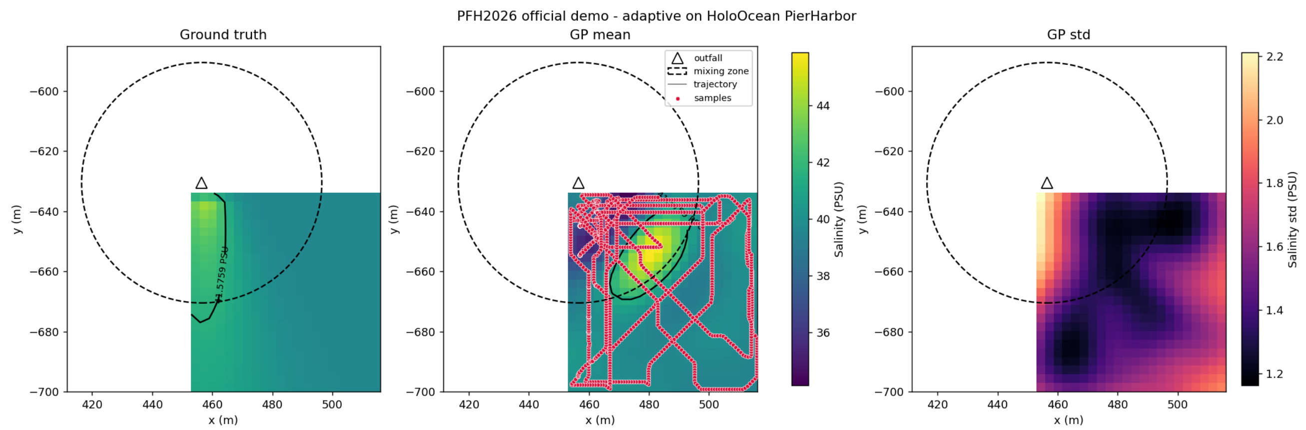
Honest reading. Adaptive sampling is most useful when the available survey distance cannot densely cover the area (-11. . . -15 % RMSE, +17 % boundary F1 at 25–50 % budget); once the budget saturates coverage, the lawnmower matches or wins. Across all 320 evaluations of both families, the three-state screening produced **zero wrong conclusive results** — every error of the legacy binary verdict became an honest REVIEW.

4.2 Sonar evidence and its limits



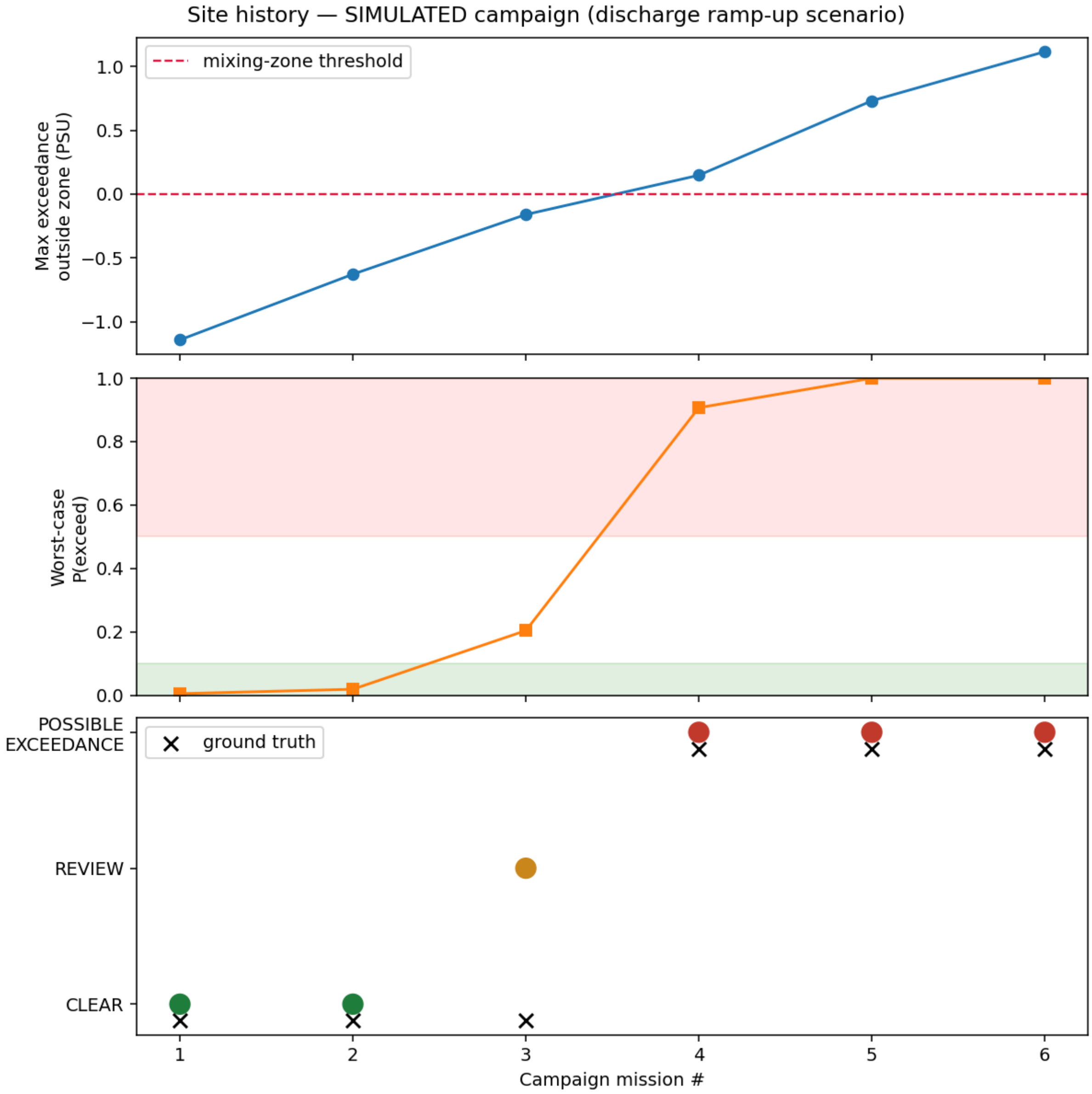
Left: on 538 recorded noisy frames, contact-strength distributions of real structure vs clutter overlap — measured, which is why single-ping gating was abandoned for persistence-based localization. The present/absent octree experiment and the open-water zero-control are in the repository (`docs/application/pfh2026/assets/sonar/`), together with the recorded-frame test fixtures that keep the perception suite running without a simulator.

4.3 The official demonstration mission



[Official HoloOcean mission (PierHarbor, BlueROV2): analytic ground truth vs GP reconstruction (trajectory and samples overlaid) vs posterior uncertainty. Sonar-confirmed localization error 6.4 m; screening: POSSIBLE EXCEEDANCE — correct for this scenario.]

4.4 Longitudinal screening (simulated campaign)



[Six simulated missions over a discharge ramp-up: the screening record tracks CLEAR → REVIEW → POSSIBLE EXCEEDANCE, with ground truth marked. Clearly labelled simulated data.]

Reproducibility. `python -m pytest tests -q` (86 tests, no GPU needed) · `python scripts/run_mission.py` (kinematic demo) · `python scripts/run_benchmark.py -seeds 20 -seed-start 100` · `python scripts/run_pfh2026_` (official HoloOcean mission). Repository: `https://github.com/AndreaBedei1/DubaiProposal`

5 Impact

Who uses it. Utilities and plant operators (feedback instead of biannual snapshots), municipalities and environmental regulators (affordable verification of mixing-zone conditions), and university or NGO laboratories in desalination-dependent regions — the users who cannot charter survey vessels routinely. **Why repeated low-cost surveys matter.** Brine impact is chronic, not episodic: a single yearly snapshot cannot separate tides, seasons and plant load. A mini-ROV mission costs a morning and a battery; the screening record turns repetition into evidence — and the REVIEW state tells operators exactly when a survey was insufficient rather than pretending certainty.

6 Roadmap

1. **Controlled water test (one day, protocol written):** CT probe calibration, response-time test, salinity-gradient pool survey, GP map vs reference casts — first non-simulated entry in the site history.
2. **Harbour trial:** a known freshwater inflow as a safe, permit-free salinity anomaly; MAVLink backend replaces the simulator backend (the mission layer is unchanged by design).
3. **Real outfall pilot** with a utility or regulator partner: site-specific mixing-zone configuration, comparison against accredited vessel casts.

7 Limitations (complete ledger in the repository)

Analytic plume surrogate, not CFD; virtual water properties; simulated sonar-to-real-sonar domain gap (thresholds must be re-tuned on real acoustics); idealized navigation; screening is *not* regulatory certification; no field validation yet; single vehicle; demo mixing zone scaled to the simulator world (all radii configurable).

8 Team

Andrea Bedei — PhD student, University of Bologna (MSc Automotive Engineering for Intelligent Mobility, 2025); supervisors Giovanni Pau and Roberto Girau, University of Bologna. Industrial support: MBS (marine robotics). Hardware available to the team: BlueROV2, Cerulean Omniscan SS450.

All figures in this document are genuine artifacts of the public repository: official HoloOcean simulator captures or matplotlib renders of recorded run data. Simulated content is labelled as such. BrineWatch is an initial prototype; it does not claim field validation, certification or commercial readiness.